

Mingbei (Tommy) Liu

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Education

B.S., Computer Science and Mathematics - Harvey Mudd College, Claremont, CA Aug 2022 – May 2026

- GPA: 3.948/4.000; Graduated with High Distinction; Honors in both Computer Science and Mathematics
- **Coursework:** Machine/Reinforcement Learning, Computer Vision, Computer Engineering, Operations Research

Languages and Technologies

Languages: Fluent in Mandarin and English; Intermediate proficiency in Spanish

Programming Languages: Python, Java, C++, C, R, MATLAB, AMPL (for linear programming), Julia, Racket

Experience

Application Development Engineer Intern, Lenovo (Beijing) Limited – Haidian, Beijing Jun 2025 – Aug 2025

- Contributed to early-stage R&D of the app store and Legion Zone software push feature; developed accurate detection and delivery of software download recommendations based on users' browser search behavior.
- Constructed training datasets and fine-tuned the BERT model; performed distillation training on the TinyBERT model to enable memory-efficient offline inference through model quantization.
- Developed a user-friendly, integrated web platform that streamlined dataset construction, model training, quantization, and evaluation, enabling efficient collaborative data labeling and model optimization.

Student R&D Engineer (Spring Team Lead), Harvey Mudd College - Claremont, CA Aug 2025 – May 2026

- Harvey Mudd College Clinic Program (Capstone Project) with Ellison Medical Institute, Los Angeles, CA
- Utilized RFID technology and reference tag localization to enable real-time tracking of samples and personnel within the clinic. Using the location data, developed a recommendation system that optimizes clinic workflow.
- Calibrated the received signal strength to physical distance, providing insights into the weighting factors for reference tag localization and enabling accurate functionality in complex environments.
- Designed an anomaly detection system to identify stagnation, loss, and process deviations using real-time location data, thereby generating actionable recommendations to optimize clinic workflows.

Student Mentor, Pomona College and Harvey Mudd College – Claremont, CA Aug 2024 – May 2026

- Mentored 7 math/CS courses (15–35 students each); delivered mini-lectures, facilitated discussions, provided assignment guidance, and supported students with write-ups. Average workload: 4 hours/course/week.
 - Calculus I (FA 2025 & SP 2026)
 - Linear Algebra (FA 2024)
 - Differential Equations (SP 2025)
 - Computability and Logic (SP 2026)
 - Mathematical Analysis (SP 2026)
 - Advanced Linear Algebra (SP 2025)
 - Operations Research (SP&FA 2025 & SP 2026)

Core Curriculum Tutor, Harvey Mudd College – Claremont, CA Jan 2025 – Dec 2025

- Tutored 3 students one-on-one (2 hours/week each) in Mechanics, General Chemistry, and Linear Algebra.

Projects

Core Algorithms Implementation

- **Machine Learning:** Decision Trees, KNN, Perceptron, Regression models, Neural Networks, Multiclass Classifiers
- **Reinforcement Learning:** Monte Carlo Control, Sarsa, Q-learning, Multi-step Temporal Difference methods

Office Hours Scheduling Optimization

- Developed a linear programming model to optimize professors' office hour schedules.
- Designed multidimensional objectives and performed sensitivity analysis on key parameters.

Nutritional Optimization of Peanut Butter Recipes

- Designed ingredient dosage models based on nutritional data to formulate optimized peanut butter recipes.
- Developed multiple models leveraging market research to meet diverse formulation objectives.